Proof of failure of strong EFM for nonadditive chores and cake

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11 October 2026

Abstract

Strong envy-freeness for mixed resources need not exist for doubly monotone item valuations. We give six agents, seven chores, and identical constant cake densities for which no complete allocation satisfies the strong EFM definition of Aziz, Lu, Mackenzie, and Suzuki. Every item has strictly negative marginal value. For the same item valuations, we determine the exact set of feasible cake totals: $[0,1]\cup[2,\infty)$. The nonexistence proof covers every item allocation and every measurable cake partition. The distinction between strong and weak EFM is essential.

1 Definition and statement

Aziz, Lu, Mackenzie, and Suzuki prove existence of EFM allocations for additive items and desirable cake, and ask whether existence extends to doubly monotone valuations [1, Section 5]. We use their Definition 2.3, including its item-only deletion condition. This is the strong EFM convention in the catalogue problem D63-72.

An allocation consists of item bundles $(O_i)_{i\in N}$ partitioning a finite set M, and measurable cake pieces $(D_i)_{i\in N}$ partitioning $[0,1]$ up to null sets. Agent i has utility

$$u_i(O,D) = v_i(O) + V_i(D), \qquad V_i(D) = \int_D f_i(t)\,dt.$$

The original question quantifies over all finite N, M, all normalized item values $v_i(\varnothing) = 0$, and all nonnegative Lebesgue-integrable f_i. It assumes that each agent has a fixed partition $M = G_i \sqcup B_i$ such that, for every $S \subseteq M$ and $g \notin S$,

$$g \in G_i \implies v_i(S\cup\{g\}) \geq v_i(S), \qquad g \in B_i \implies v_i(S\cup\{g\}) \leq v_i(S).$$

The sign partition may depend on the agent. Such values are called *doubly monotone*; either part may be empty. An allocation is *strong EFM* if, for every ordered pair i, j, either

$$v_i(O_i) + V_i(D_i) \geq v_i(O_j) + V_i(D_j), \tag{1}$$

or

$$V_i(D_j) = 0 \quad \text{and} \quad \exists g \in O_i \cup O_j : \quad v_i(O_i \setminus \{g\}) \geq v_i(O_j \setminus \{g\}). \tag{2}$$

In particular, the last inequality does not include $V_i(D_i)$.

Let

$$N = \{X_1, X_2, Y_1, Y_2, Z_1, Z_2\}, \qquad M = \{x, y, z, d_1, d_2, d_3, d_4\}.$$

The designated item a_i is x for agents of type X, y for type Y, and z for type Z. Define $v_i = -c_i$, where, for every $S \subseteq M$,

$$c_i(S) = \begin{cases} 0, & S = \varnothing, \\ 1, & S = \{a_i\}, \\ 2, & |S| = 1 \text{ and } a_i \notin S, \\ 5, & |S| = 2 \text{ and } a_i \in S, \\ 3, & |S| = 2 \text{ and } a_i \notin S, \\ |S| + 3, & |S| \geq 3. \end{cases} \tag{3}$$

All agents have the same constant density $f_i(t) = C$, where $C \geq 0$.

Theorem 1. *For the valuations* (3)*, a complete strong-EFM allocation exists if and only if*

$$C \in [0, 1] \cup [2, \infty).$$

Consequently, taking $C = 3/2$ gives a counterexample to universal strong-EFM existence for doubly monotone items and nonnegative integrable cake densities.

The instance satisfies the original assumptions. Every cost is normalized, and adding any item strictly increases cost: the increases from sizes 0 to 1, 1 to 2, and 2 to 3 are at least one; all subsequent increases equal one. Thus each v_i is strictly decreasing under every item addition. It is doubly monotone with $G_i = \varnothing$ and $B_i = M$. The costs are nonadditive: for example, $c_{X_1}(\{x, y\}) = 5 > c_{X_1}(\{x\}) + c_{X_1}(\{y\}) = 3$.

For any cake partition write

$$p_j = C\,|D_j|, \qquad p_j \geq 0, \qquad \sum_j p_j = C, \tag{4}$$

where $|D_j|$ denotes Lebesgue measure. Every agent values D_j at the same number p_j. Conversely, for $C > 0$ every nonnegative vector with sum C is realized by consecutive cake intervals of lengths p_j/C. For $C = 0$, every cake partition has value zero.

2 Nonexistence for $1 < C < 2$

Suppose a strong-EFM allocation exists, and use (4).

Every item bundle is nonempty. If $O_j = \varnothing$, some other bundle O_i contains at least two of the seven items. Its cost to its owner is at least three, and hence

$$-c_i(O_i) + p_i \leq -3 + C < 0 \leq p_j.$$

Agent i envies j. Deleting any one item from O_i leaves a nonempty bundle of positive cost, whereas O_j is empty and has no item to delete. The item-only inequality in (2) therefore fails. This excludes every allocation with an empty bundle.

The unique pair contains its owner's designated item. Seven items in six nonempty bundles give one pair $P = O_h$ and five singletons. If $a_h \notin P$, some agent j holds $\{a_h\}$. Then

$$-c_h(P) + p_h \leq -3 + C < -1 \leq -c_h(\{a_h\}) + p_j.$$

Deleting either item from P leaves cost two, exceeding the compared cost one. Deleting the compared item instead leaves compared cost zero and own cost three. These exhaust $P \cup \{a_h\}$, so no deletion satisfies (2). Necessarily $a_h \in P$, and $c_h(P) = 5$.

Only the pair owner can receive cake. For every singleton owner $j \neq h$, its item costs at most two to h. Consequently

$$-c_h(P) + p_h \leq -5 + C < -3 < -2 \leq -c_h(O_j) + p_j.$$

Thus h envies every other agent. Condition (2) requires $p_j = 0$ for each $j \neq h$, and completeness gives $p_h = C$.

An omitted designated item forces a contradiction. The pair P omits at least one of x, y, z; denote an omitted item by z, relabeling the three types if necessary. Both type-Z agents are singleton owners, since the pair contains its owner's designated item. At most one of them holds $\{z\}$. Choose the other, say k. Its own cost is two, whereas its cost for P is three. Therefore

$$-c_k(O_k) + p_k = -2 < -3 + C = -c_k(P) + p_h.$$

Agent k envies the pair owner, whose cake has positive value. Neither (1) nor (2) can hold. This contradiction proves nonexistence throughout $(1, 2)$.

3 The remaining cake totals

For $0 \leq C \leq 1$, give $\{x, y\}$ and all the cake to X_1, give $\{z\}$ to Z_1, and assign d_1, d_2, d_3, d_4 singly to X_2, Y_1, Y_2, Z_2, respectively. Every singleton owner can remove its own item to obtain item cost zero, so comparisons toward cake-free bundles satisfy (1) or (2). The pair owner can retain x, of cost one, after deleting y; this also handles all of its comparisons toward cake-free bundles. No singleton owner envies the paid pair: the three remaining X/Y agents have own cost two and pair cost five, Z_1 has own cost one and pair cost three, and Z_2 has own cost two and pair cost three. The inequalities $2 \leq 5 - C$, $1 \leq 3 - C$, and $2 \leq 3 - C$ hold for $C \leq 1$. Hence the allocation is strong EFM, including both endpoints.

For $C = 2$, give $\{x, y\}$ and all cake to Z_1, give $\{z\}$ to Z_2, and assign the ordinary chores singly to the four X/Y agents. This allocation is fully envy-free. Agent Z_1 has net cost one, tying $\{z\}$ and preferring its package to each ordinary singleton. Agent Z_2 has cost one and values the paid pair at net cost one. Every X/Y agent has cost two, values every singleton at cost two, and values the paid pair at net cost three. For $C > 2$, use the same item allocation and payments

$$p_{Z_1} = 2 + \frac{C-2}{6}, \qquad p_i = \frac{C-2}{6} \quad (i \neq Z_1).$$

Adding the same amount to every payment preserves full envy-freeness. Equation (4) realizes these payments as a complete cake partition. This completes the proof of Theorem 1.

4 Why the strong definition matters

Remark 2.4 of [1] distinguishes a weaker condition that retains the owner's cake value in the deletion comparison. Our counterexample does not establish failure of that weaker condition. Indeed, for $1 < C < 2$, use the last section's $C = 2$ item allocation and give all cake to Z_1. Only Z_1 envies Z_2: its net cost is $3 - C > 1$. After deleting one of its own two items, the weak comparison has net cost $2 - C < 1$, so it passes. Every other comparison is already envy-free. The strong item-only comparison instead has cost two against cost one and fails. Thus the negative answer applies precisely to the stronger definition specified in D63-72.

5 Scope

Theorem 1 gives a negative answer to the universal existence question in D63-72 and in Section 5 of [1], under Definition 2.3. A single admissible instance without an allocation suffices to refute that assertion. The additional endpoint constructions establish the exact budget set for this instance. There is no external mathematical theorem on which the proof depends.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] Haris Aziz, Xinhang Lu, Simon Mackenzie, and Mashbat Suzuki. *Fair Division with Indivisible Goods, Chores, and Cake*. arXiv:2511.04891v1, 7 November 2025. Definition 2.3, Remark 2.4, and Section 5. `https://arxiv.org/abs/2511.04891v1`.